

High-Speed ATM Transaction Switch Processing System

Domain: Financial Domain (ATM Security and Transactions) | Statutory Architecture

Parameter	Evaluation Result	Governance & Statutory Impact
MoSCoW Baseline	Must Have	Ensures critical functionalities that comply with stringent regulations and maintain operational integrity.
Risk Classification	CRITICAL (98/100)	Mission-critical financial tier. Direct automated capital risk or physical cash exposure requiring zero-trust stage gates.
Total Requirements	41 Modular Requirements (Min 30)	Decomposed across Frontend, Backend, Auth, Security, Database Residency, and Resilience

1. System Scope & Stakeholder Requirement Statement

"High-speed ATM transaction switch processing ISO 8583 messages, interfacing with FIPS 140-3 HSM, handling vault door reed sensors and cash cassette dispensers with automated zeroization on physical tamper."

2. Modular Software Requirements Specification (41 Detailed Points)

Pillar: Frontend & Mobile UI/UX (5 Specifications)

Req ID	Title & Detailed Specification	Priority	Statutory Citation & Acceptance Criteria
REQ-FE-01	ATM Touchscreen Interactive Customer Interface Deliver high-contrast, accessible touch interface for ATM users of High-Speed ATM Transaction Switch Processing System with multi-language voice guidance and dynamic currency denomination selection.	Must Have	Regulation: RBI Master Circular on Customer Service in Banks & ATM Accessibility Ref: Atm Switching Network Software Requirements Specification Ieee830 Acceptance: Renders touch interaction with response time < 150ms; supports 12 regional languages.
REQ-FE-02	Technician Maintenance & Cassette Diagnostics Console Provide secure diagnostic interface accessible only via authenticated maintenance smartcards to inspect cash levels, bill validator health, and mechanical jams.	Must Have	Regulation: RBI Cyber Security Controls for Third Party ATM Switch ASPs Ref: Rbi Atm Security Guidelines Acceptance: Enforces session timeout after 5 minutes of idle time; logs all diagnostic operations.
REQ-FE-03	Real-Time Cash Cassette Capacity & Status Dashboard Provide live visual dashboard displaying current note counts, denomination distribution, low-cash warnings, and cash run-out forecast per cassette.	Should Have	Regulation: RBI Guidelines on Cash Replenishment Operations Acceptance: Updates cassette status in real time upon every cash dispense and replenishment.
REQ-FE-04	Vault Door & Physical Tamper Alert Console Display instant visual and acoustic alerts for security staff upon unauthorized vault opening, vibration sensor triggers, or physical tilt detection.	Must Have	Regulation: RBI Security Measures for ATMs & Currency Chests Acceptance: Triggers emergency alert on operations dashboard within 2 seconds of sensor event.
REQ-FE-05	Branch Manager Remote Oversight & Incident Management Portal Provide web-based management portal for bank branch managers to monitor ATM network uptime, cash replenishment cycles, and technician dispatch.	Should Have	Regulation: RBI Business Continuity Planning for Retail Banking Acceptance: Generates daily ATM cash reconciliation and uptime SLA compliance reports.

Pillar: Backend Architecture & ATM Switching (6 Specifications)

Req ID	Title & Detailed Specification	Priority	Statutory Citation & Acceptance Criteria
REQ-BE-01	ISO 8583 Message Handling Processes ISO 8583 messages for ATM transactions securely.	Should Have	Regulation: FIPSPolicy Acceptance: Successfully handles at least 99% of ISO 8583 messages without errors.
REQ-BE-02	IoT Sensor Telemetry & Event Ingestion Pipeline Deploy high-throughput IoT daemon ingesting telemetry from cassette sensors, door reed switches, vibration gyroscopes, and thermal sensors via mTLS.	Must Have	Regulation: RBI Cyber Security Framework for ATM Switch ASPs Ref: Atm Switching Network Software Requirements Specification Ieee830 Acceptance: Processes 5,000 sensor telemetry events/second with guaranteed at-least-once delivery.
REQ-BE-03	Cash Cassette Depletion Tracking & Automated Forecasting Engine Deploy analytics engine calculating real-time burn rates per denomination, triggering automated cash transit replenishment orders before stock-out.	Must Have	Regulation: RBI Currency Management & Clean Note Policy Guidelines Ref: Rbi Atm Security Guidelines Acceptance: Dispatches cash replenishment order when cassette falls below 15% threshold.
REQ-BE-04	ATM Transaction Switch Engine & ISO 8583 Daemon Process ATM cash withdrawal, deposit, and balance inquiry messages using ISO 8583 financial message standards with cryptographic MAC validation.	Must Have	Regulation: NPCI National Financial Switch (NFS) Operating Guidelines Acceptance: Maintains end-to-end transaction latency under 800ms with 99.99% switch uptime.
REQ-BE-05	Automated Electronic Vault Lock Management Microservice Enforce dynamic one-time password (OTP) and dual-authorization codes required to open electronic safe locks during cash replenishment.	Must Have	Regulation: RBI Directives on Cassette Swap in ATMs (RBI/2018-19/166) Acceptance: Generates time-bound (15-minute) dynamic vault lock codes with dual authorization.
REQ-BE-06	Emergency Incident Alert & Siren Dispatcher Deploy automated notification engine dispatching instant SMS, push, and automated IVR phone alerts to the nearest police station and branch security upon tamper.	Must Have	Regulation: RBI Security Measures for ATMs — CCTV & Alarm Integration Acceptance: Dispatches alert notifications to security officers within 3 seconds of breach.

Pillar: Authentication, Identity & Access Governance (6 Specifications)

Req ID	Title & Detailed Specification	Priority	Statutory Citation & Acceptance Criteria
REQ-AUTH-01	User Authentication and Authorization Enforces user authentication and authorization based on predefined roles and permissions.	Should Have	Regulation: RBI Regulations Acceptance: Validates at least 5000 users per hour, ensuring correct role-based access control.
REQ-AUTH-02	Dual-Custody Biometric Authentication for Vault Opening Require two authorized personnel (Bank Official + Cash Transit Agent) to simultaneously present biometric credentials to unlock ATM vault cassettes.	Must Have	Regulation: RBI Directives on Cash Logistics & ATM Vault Security Ref: Atm Switching Network Software Requirements Specification Ieee830 Acceptance: Prohibits single-person vault access; requires dual biometric confirmation.
REQ-AUTH-03	ATM Cardholder EMV Chip & PIN Authentication Process EMV Level 2 chip transactions with dynamic cryptogram verification and encrypted PIN block translation via Hardware Security Module (HSM).	Must Have	Regulation: NPCI NFS Chip & PIN Migration Mandate Ref: Rbi Atm Security Guidelines Acceptance: Rejects fallback to magnetic stripe; validates ARQC cryptogram with issuing bank.

REQ-AUTH-04	Technician Smartcard & Hardware Token Authorization Authenticate field engineers using PKI-based hardware cryptographic tokens before granting access to maintenance menus and USB ports.	Must Have	Regulation: RBI Circular on Cyber Security Controls for Third Party ATM Switch ASPs Acceptance: <i>Blocks USB peripheral access without active authenticated technician session.</i>
REQ-AUTH-05	Central Switch Terminal Master Key (TMK) Remote Injection Support secure Remote Key Loading (RKL) of Terminal Master Keys (TMK) and PIN Encryption Keys (PEK) into the Encrypting PIN Pad (EPP).	Must Have	Regulation: PCI PTS (PIN Transaction Security) v5.0 Standard Acceptance: <i>Keys injected over TLS 1.3 encrypted channel directly into tamper-responsive EPP.</i>
REQ-AUTH-06	Audit Session Binding for Maintenance Operations Bind all maintenance actions to technician employee ID, geotagged location, and video surveillance timestamp with dual sign-off.	Must Have	Regulation: RBI IT Governance & ATM Physical Security Guidelines Acceptance: <i>Logs technician entry and exit timestamps with matched CCTV snapshot reference.</i>

Pillar: Information Security & Physical Controls (7 Specifications)

Req ID	Title & Detailed Specification	Priority	Statutory Citation & Acceptance Criteria
REQ-SEC-01	Interface with FIPS 140-3 HSM Interacts with FIPS 140-3 certified Hardware Security Module (HSM) for cryptographic operations.	Must Have	Regulation: FIPSPolicy Acceptance: <i>Passes FIPS 140-3 certification audit.</i>
REQ-SEC-02	Dual-Custody Electronic Vault Lock Generation Generates dynamic, time-limited electronic vault locks for cash replenishment under dual-custody supervision.	Must Have	Regulation: RBI Regulations Acceptance: <i>Generates at least 1000 unique locks per day, ensuring each lock is valid for exactly 15 minutes.</i>
REQ-SEC-03	Cryptographic Key Zeroization Automatically zeroes out cryptographic keys when necessary to ensure data protection.	Must Have	Regulation: FIPSPolicy Acceptance: <i>Zeroes out cryptographic keys within 0.5 seconds after receiving a zeroize command.</i>
REQ-SEC-04	RBI Security Measures for ATMs & Currency Chests Compliance Implement comprehensive technical safeguards complying with RBI Master Direction on physical, cyber, and operational security of ATMs.	Must Have	Regulation: RBI/2018-19/166 DCM (Plg) No.2968/10.25.007/2018-19 Ref: Atm Switching Network Software Requirements Specification leee830 Acceptance: <i>Enforces cassette swap architecture, vault bolting to floor, and anti-skimming.</i>
REQ-SEC-05	Hardware Security Module (HSM) PIN Encryption & Translation Encrypt all PIN entry blocks inside FIPS 140-2 Level 3 certified Encrypting PIN Pads (EPP) using Triple-DES / AES-128 before memory transmission.	Must Have	Regulation: PCI PTS & RBI Digital Payment Security Controls Ref: Rbi Atm Security Guidelines Acceptance: <i>Zero plaintext PIN persistence in ATM memory, OS storage, or transmission lines.</i>
REQ-SEC-06	Tamper-Evident Sensor System & Hardened Vault Alarms Equip cash vault with seismic vibration sensors, thermal rate-of-rise detectors, and door intrusion sensors triggering local sirens and central alarms.	Must Have	Regulation: RBI ATM Physical Security Controls Directive Acceptance: <i>Triggers siren and central alert immediately upon physical attack or tilt.</i>
REQ-SEC-07	Operating System Hardening & USB Port Whitelisting Apply rigorous OS-level security policies disabling autorun, enforcing application whitelisting, and disabling all unused physical USB and Ethernet ports.	Must Have	Regulation: RBI Cyber Security Controls for ATM Switch ASPs Acceptance: <i>Blocks unauthorized executable execution; prevents black-box malware attacks.</i>

Pillar: Database Architecture & Telemetry Residency (5 Specifications)

Req ID	Title & Detailed Specification	Priority	Statutory Citation & Acceptance Criteria
REQ-DB-01	100% Indian Sovereign Data Residency for Telemetry & Logs Store all ATM sensor telemetry, transaction logs, and surveillance metadata exclusively inside physical data centers located on Indian soil.	Must Have	Regulation: RBI Storage of Payment System Data Directive Ref: Atm Switching Network Software Requirements Specification Ieee830 Acceptance: <i>Certifies zero foreign storage of ATM operational or transaction records.</i>
REQ-DB-02	High-Volume Time-Series Telemetry & Sensor Event Store Deploy optimized time-series database storing sensor readings (temperature, cash counts, vibration metrics) with 1-second resolution.	Must Have	Regulation: RBI Operational Resilience Guidelines Ref: Rbi Atm Security Guidelines Acceptance: <i>Sustains 10,000 sensor metrics/second; supports rapid anomaly pattern search.</i>
REQ-DB-03	Double-Entry Cash Dispensing Audit Ledger Maintain relational financial ledger matching cassette physical cash deductions against terminal electronic journals and host settlement messages.	Must Have	Regulation: RBI Currency Management & Accounting Norms Acceptance: <i>Identifies cash overage/shortage discrepancies within T+0 end-of-day settlement.</i>
REQ-DB-04	Statutory 10-Year Hardware Incident & Audit Trail Archival Automate cold storage archival of all ATM door openings, cassette replacements, and maintenance events with cryptographic hash chains.	Must Have	Regulation: PMLA 2002 & RBI Incident Reporting Guidelines Acceptance: <i>Preserves tamper-proof maintenance records with annual integrity verification.</i>
REQ-DB-05	High-Availability Database Clustering & Host Failover Deploy active-standby database replication for the ATM switch management cluster ensuring seamless failover without transaction interruption.	Should Have	Regulation: RBI Business Continuity Planning Framework Acceptance: <i>Failover completes in <60 seconds with zero committed transaction log loss.</i>

Pillar: Resilience, Scalability & Disaster Recovery (6 Specifications)

Req ID	Title & Detailed Specification	Priority	Statutory Citation & Acceptance Criteria
REQ-RES-01	Failover Mechanism for Critical Components Provides failover mechanisms for critical components like HSM and NFS interface to ensure uninterrupted service.	Should Have	Regulation: FIPSPolicy Acceptance: <i>Reduces downtime due to component failures by at least 50%.</i>
REQ-RES-02	Offline Cash Dispense Reconciliation & Reconnect Sync Enable ATM safe recovery mode during communication link drop, safely rolling back uncompleted transactions and syncing state upon link restoration.	Must Have	Regulation: NPCI NFS Operating Guidelines on Offline Recovery Ref: Atm Switching Network Software Requirements Specification Ieee830 Acceptance: <i>Prevents balance debit without cash dispensation; synchronizes journal on reconnect.</i>
REQ-RES-03	Dual-SIM Cellular & Redundant Leased Line Network Uplink Equip ATM terminal with dual-SIM 4G/5G cellular routers and wired leased line with automated carrier failover upon packet loss > 5%.	Should Have	Regulation: RBI ATM Network Availability Norms Ref: Rbi Atm Security Guidelines Acceptance: <i>Switches to secondary carrier in <5 seconds without terminating active user session.</i>
REQ-RES-04	UPS Battery Backup Monitoring & Graceful Emergency Shutdown Integrate smart uninterruptible power supply (UPS) monitoring battery charge and triggering orderly transaction rollback and system shutdown before battery drain.	Must Have	Regulation: RBI IT Infrastructure & Operational Resilience Directives Acceptance: <i>Guarantees minimum 4-hour battery run-time; safe shutdown prevents DB corruption.</i>
REQ-RES-05	Automated Switch Failover to Backup Core Banking Host Configure multi-homed ATM switch routing traffic to secondary core banking switch upon primary host response latency > 3 seconds.	Must Have	Regulation: RBI Harmonisation of TAT & Customer Compensation Directives Acceptance: <i>Redirects traffic seamlessly; ensures 99.95% annual ATM service availability.</i>

REQ-RES-06	Severe Tamper Auto-Lockdown & Cryptographic Zeroization Incorporate cryptographic zeroization circuitry that instantly wipes Terminal Master Keys and locks physical cash cassettes upon extreme physical breach.	Must Have	Regulation: PCI PTS & Global ATM Security Standards Acceptance: Zeroizes cryptographic keys in <100 milliseconds upon physical vault penetration.
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Pillar: Hardware Interfaces (1 Specifications)

Req ID	Title & Detailed Specification	Priority	Statutory Citation & Acceptance Criteria
REQ-EXT-36	Vault Door Reed Sensor Interfacing Interfaces with vault door reed sensors to monitor physical access to the vault.	Must Have	Regulation: RBI Regulations Acceptance: Correctly detects open/closed status of vault doors with no false positives.

Pillar: Operational Functions (2 Specifications)

Req ID	Title & Detailed Specification	Priority	Statutory Citation & Acceptance Criteria
REQ-EXT-37	Cash Cassette Dispenser Automation Automatically zeros out cash cassette dispensers upon detection of physical tampering.	Must Have	Regulation: RBI Regulations Acceptance: Zeroes out cash cassette dispensers within 1 second after detecting tampering.
REQ-EXT-39	Real-Time Monitoring of ATM Peripherals Monitors real-time operation of ATM peripherals such as card readers, pin pads, and bill validators.	Should Have	Regulation: FIPSPolicy Acceptance: Detects and alerts on any malfunctioning peripheral devices within 3 seconds.

Pillar: Integration (1 Specifications)

Req ID	Title & Detailed Specification	Priority	Statutory Citation & Acceptance Criteria
REQ-EXT-38	Secure Communication with NFS Establishes secure communication with the National Financial Switch (NFS) through NDC+ protocol.	Should Have	Regulation: FIPSPolicy Acceptance: Maintains secure communication with NFS for at least 95% of transactions.

Pillar: Logging and Auditing (1 Specifications)

Req ID	Title & Detailed Specification	Priority	Statutory Citation & Acceptance Criteria
REQ-EXT-40	Audit Trail Logging Records detailed logs of all ATM transactions and system events for auditing purposes.	Should Have	Regulation: RBI Regulations Acceptance: Logs at least 1000 events per minute, including timestamps and event types.

Pillar: Encryption (1 Specifications)

Req ID	Title & Detailed Specification	Priority	Statutory Citation & Acceptance Criteria
REQ-EXT-41	Data Encryption and Decryption Encrypts and decrypts sensitive data using industry-standard encryption algorithms.	Should Have	Regulation: FIPSPolicy Acceptance: Handles at least 10000 encrypted transactions per day, ensuring data confidentiality.

3. Regulatory & Statutory Grounding (Indian Banking Frameworks)

Framework	Enforcement Level	Statutory Directive / Citation
RBI Security Measures for ATMs & Currency Chests	Physical & Sensor Vault Controls	RBI/2018-19/166 DCM (Plg) No.2968/10.25.007/2018-19
RBI Cyber Security Controls for Third Party ATM Switch ASPs	ATM OS & Hardware Whitelisting	RBI/2021-22/68 DOS.CO.CSITE.SEC.No.1852/31.01.015/2021-22
NPCI National Financial Switch (NFS) Guidelines	ISO 8583 Message Standards	NPCI NFS Procedural Guidelines v4.1
PCI PTS (PIN Transaction Security) v5.0	Encrypting PIN Pad (EPP) Security	PCI Security Standards Council PTS Directive

4. Mandatory Security & Architectural Controls

Control ID	Title & Architecture Specification	Automated Verification Method
CTRL-SEC-01	Financial Cryptographic Transport & Rest Encryption Enforce TLS 1.3 for all in-transit communications and AES-256-GCM for sensitive fields.	Automated security scanning and cryptographic key rotation verification.
CTRL-AUTH-02	Dual-Factor Authentication & Device Fingerprint Binding Mandatory 2FA with device hardware fingerprint binding and session timeout enforcement.	Penetration testing verifying zero session hijack or credential bypass.
CTRL-AUD-03	Statutory Immutable Forensic Audit Logging Maintain append-only WORM audit logs with microsecond NTP timestamps for statutory retention period.	Cryptographic checksum validation & CERT-In audit verification.
CTRL-RES-04	Automated Circuit Breaker & High Availability Failover Deploy Resilience4j circuit breakers and active-standby database failover with RPO < 1m.	Live chaos drill and automated failover switchover test.

5. Traceability Matrix & Recommended SDLC Sprints

Req IDs	Target Sub-Module	SDLC Phase / Gate	Verification Deliverable
REQ-FE-01 - REQ-FE-05	High-Speed ATM Transaction Switch Processing System — Frontend & Mobile UI/UX	Sprint 1 / Frontend Milestone	Frontend & Mobile UI/UX Architecture & Verification Suite <i>Automated Integration Test & Security Signoff</i>
REQ-BE-01 - REQ-BE-06	High-Speed ATM Transaction Switch Processing System — Backend Architecture & ATM Switching	Sprint 2 / Backend Milestone	Backend Architecture & ATM Switching Architecture & Verification Suite <i>Automated Integration Test & Security Signoff</i>
REQ-AUTH-01 - REQ-AUTH-06	High-Speed ATM Transaction Switch Processing System — Authentication, Identity & Access Governance	Sprint 3 / Authentication, Milestone	Authentication, Identity & Access Governance Architecture & Verification Suite <i>Automated Integration Test & Security Signoff</i>
REQ-SEC-01 - REQ-SEC-07	High-Speed ATM Transaction Switch Processing System — Information Security & Physical Controls	Sprint 4 / Information Milestone	Information Security & Physical Controls Architecture & Verification Suite <i>Automated Integration Test & Security Signoff</i>
REQ-DB-01 - REQ-DB-05	High-Speed ATM Transaction Switch Processing System — Database Architecture & Telemetry Residency	Sprint 5 / Database Milestone	Database Architecture & Telemetry Residency Architecture & Verification Suite <i>Automated Integration Test & Security Signoff</i>

REQ-RES-01 - REQ-RES-06	High-Speed ATM Transaction Switch Processing System — Resilience, Scalability & Disaster Recovery	Sprint 6 / Resilience, Milestone	Resilience, Scalability & Disaster Recovery Architecture & Verification Suite <i>Automated Integration Test & Security Signoff</i>
REQ-EXT-36	High-Speed ATM Transaction Switch Processing System — Hardware Interfaces	Sprint 7 / Hardware Milestone	Hardware Interfaces Architecture & Verification Suite <i>Automated Integration Test & Security Signoff</i>
REQ-EXT-37 - REQ-EXT-39	High-Speed ATM Transaction Switch Processing System — Operational Functions	Sprint 8 / Operational Milestone	Operational Functions Architecture & Verification Suite <i>Automated Integration Test & Security Signoff</i>
REQ-EXT-38	High-Speed ATM Transaction Switch Processing System — Integration	Sprint 9 / Integration Milestone	Integration Architecture & Verification Suite <i>Automated Integration Test & Security Signoff</i>
REQ-EXT-40	High-Speed ATM Transaction Switch Processing System — Logging and Auditing	Sprint 10 / Logging Milestone	Logging and Auditing Architecture & Verification Suite <i>Automated Integration Test & Security Signoff</i>
REQ-EXT-41	High-Speed ATM Transaction Switch Processing System — Encryption	Sprint 11 / Encryption Milestone	Encryption Architecture & Verification Suite <i>Automated Integration Test & Security Signoff</i>